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ORIGINAL RESEARCH

A Multilayered Encryption for the Robust Design and Implementation of Secure Audio Files

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Abstract

The proliferation of digital technologies has resulted in the creation of a great deal of cryptographic techniques for the safe transfer and storage of private information. There is a widespread application of the SHA-256 technique, which is a cryptographical hash function, to verify the validity, identity, and integrity of data. In this research, a multi-layer approach to enhance the security of audio encryption has been adopted. The Exceptional features of audio files ensure a strong defense alongside potential cryptographic vulnerabilities is intended to be addressed by this methodological plan. The waveform analysis, histogram analysis, correlation analysis, spectrogram analysis, number of sample change rate referred to as NSCR, key sensitivity, universal average changing intensity referred to as UACI, peak single to noise ratio, signal to noise ratio, crest factor and values of the root-mean-square is presented. The proposed method displays favorable numerical results across different factors. The result of the correlation coefficient is 0.000119, representing a minimal linear relationship between original wav file and encrypted wav file. With a key space of $\mathbf{2^{256}}$, the algorithm ensures a robust cryptographic foundation. Additionally, it maintains a high NSCR value of **99.9985%**, indicating significant sample alteration, while offering a low UACI value of **0.33%**, suggesting minimal average intensity changes between encrypted and original audio. Based on the findings of this result analysis, it is clear that the audio encryption approach that was provided is suitable for use in applications that include secure speech transmission.

Keywords Cryptography · Security · Audio file security · Encryption · Decryption

Introduction

Due to the fact that audio transmission has become an essential component of everyday life, it is imperative that the confidentiality of private conversations be protected. In the given aspect, audio encryption techniques could be utilized to ensure the safe transfer of an audio file with great security. For the purpose of preventing unauthorized third parties from correctly decrypting the audio that is being transferred, audio samples are having encryption using a secret key which is only known to the similar side. Therefore, even if an adversary were to gain access to the encrypted data, he would be unable to comprehend the real content of the audio.

The input audio signals can be subjected to scrambling and diffusion procedures, which are approaches that are analogous to those used for image encryption [1]. The technique of scrambling involves rearranging the audio samples by utilizing a sequence that is dependent on the keys that are used. Following the process of scrambling, the diffusion operations alters the rate of the audio samples in accordance with a diffusion generation. In the body of research that has been done, many chaotic maps have been utilized to create diffusion and scrambling parameters. An instance of an audio encryption technique that is proposed in [2] is one that makes use of a three-dimensional scroll chaotic model and is based on confusion and diffusion operations.

In a similar manner, the encryption of audio messages is accomplished in [3] by utilizing a combination of three distinct chaos functions. In [4], a method for encrypting speech is described that is based on chaotic maps. These chaotic maps include Bernoulli's map, logistic map, quadratic map,

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Chen's map, and tent map. For the purpose of developing an audio encryption strategy [5], combines DNA encoding with a multi-chaotic systems and bit-scrambling process. Piecewise linear chaotic map and DNA encoding are both utilized in an audio encryption technique within [6], which is similar to the previous example. Regarding yet another approach for the encryption of digital audio signals, the Mobius transformation is utilized for the replacement process, and pixel permutation is accomplished by using the Henon map which is chaotic in nature [7]. For the purpose of encrypting audio files, a pseudorandom byte generator that is on the basis of the Ikeda chaotic map is utilized [8]. When it comes to another strategy, the Chen memristor chaotic model is utilized [9]. There are additional alternative ways of audio encryption that do not include chaos that can be found in the literature [10, 11]. There are algorithms for many different kinds of data, such as audio, video, and photographs, among others [20]. The suggested multilayer approach can be used to encrypt audio files in order to increase the security of and the audio file [23]. Latest progress in speech encryption shows substantial achievement in applications of chaotic systems and multilayered security frameworks. Abdallah and Meshoul (2023) proposed a multilayered encryption mechanism using four chaotic maps (Sine, Logistic, Tent, and Henon) along with DNA coding that provides strong security through block-wise confusion/diffusion and DNA/XNOR operations for producing final cipher [29]. It overcomes the weaknesses of single-map systems through diversification of chaotic dynamics among encryption phases. Adhikari and Karforma (2021) confirmed the strength of hybrid chaotic systems by their Henon-Tent composite model, which strengthens pseudo-random sequence complexity for better resistance to statistical attacks [30]. Comparative investigations by Haridas et al. (2024) show that 3D hyperchaotic systems are better than 2D counterparts in audio encryption with 18% higher Lyapunov exponents (1.72 vs. 1.46) and better key sensitivity indicators [31]. For speech-specific applications, Gebereselassie and Roy (2023) created customized chaotic maps optimized for vocal frequency ranges (300–3400 Hz), tested with NIST SP 800−22 tests reporting 97.3% pass rates in randomness testing [32]. These works as a whole indicate an industry trend toward multi-mechanism cryptographic integration, albeit with a need for standardized performance benchmarks for real-time voice communication systems.

Background

SHA256 Algorithm

An input message of nearly two 64 bits is processed by the SHA-256 technique, which then generates a hash of 256 bits in length that is fixed in total length. Both the message preparation and the hash computation stages are the two primary processes that make up the algorithm. During the pre-processing stages, the input messages are padded so that it is 512 bits long and then divided into blocks. A 256-bit message digest is produced as a result of the hash computation stage, which involves expanding and compressing each block through a series of arithmetic and logical operations. Collision attacks, in which two varied inputs generate similar results, and pre-image assaults, in which an attacker attempts to identify an input which generates a certain output, are two examples of the types of attacks that the algorithm is supposed to be resistant to. The method is designed to be resistant on multiple fronts.

During the pre-processing phase, the input messages is initially being encoding using the UTF-8 binary standards. As a delimiter, a bit with a value of one is inserted to the input message's end that has been encoded with a length of one. This bit indicates that the binary sequence has reached its conclusion. Then, the bounded thread is non-padded to a maximum of k bits, and a 64-bit big-endian method is employed to append the length of the first input message to the end location. This is done until the string is at the desired length. This is done for the entire string until it is finished. The lowest non-negativity solution that will make the equation $(l+1+k+64) \bmod 512 \equiv 0$ valid needs to be found in order to establish the value of k. The symbolism $(A) \bmod B \equiv 0$ signifies A is an integer multiple of B. Subsequently the message that has been padded is divided into a number of 512-bit blocks marked with the notation M(i), where i can be any number in the range of 1 and N. Size of each block is of sixteen 32-bit words denoted by the symbol Mt(i), and the value of t falls between 0 and 15. Final aspect of the pre-processing phase is the initializing of the working variables which is ultimately the final step. Hash values are stored in this variable and the same are updated during the iteration of the method. Here the variables a to h represent working variables with the initializes corresponding values HO(0) to H7(0) as per the information present in the legitimate source of documentation of FIPS 180-4 Secure Hash Standard [12]. At the end of each block, the working variables are upgraded with the middle hash value H(i) for the message blocks that follow (i=2, 3,., N). This continues for every block that comes after. The updated working variables will be used as the starting values for the subsequent block of the message till final digest represented by H(N) is

computed, after their calculation. Because of the way this system is set up, hash functions are applied to the complete messages instead of just specific blocks [12].

The SHA-256 procedure is illustrated in its entirety above in Fig. 1 [24].

ChaCha20 Encryption Algorithm

Chacha20 is an updated version of Salsa20 that takes advantage of a new round function to increase the amount of diffusion. A rotation operations dependent core hash function, XOR, and 32-bit module addition is utilized by Salsa20. Conversely, Chacha20 uses an internal state of sixteen 32-bit words arranged into a 4×4 matrix. The matrix is utilized to map a 64-bit counter to a 512-bit block key stream, a 64-bit nonce, and a 256-bit key (a 256-bit key is specified at the same time for Salsa20). Though the configurations are not the same, the initial states of the two of them contain the same number of words: four constant words, two nonce words, and two stream position (counter) words and eight key words. Salsa20's quarter round is replaced by Chacha20, which does twice as many operations on each syllable. Salsa20 has been replaced with Chacha20. In a double round, Chacha20 is performing down diagonals as well as quarter rounds down columns, in contrast to Salsa20, that is alternating quarter rounds across rows and down columns during the course of the game. Similarly, Chacha20 uses 10 repetitions with double-round and the same output function [13].

$$\left\{\begin{array}{lll} k+=x; & z\oplus=k; & z\lll=16; \\ y+=z; & x\oplus=y; & x\lll=12; \\ k+=x; & z\oplus=k; & z\lll=8; \\ y+=z; & x\oplus=y; & x\lll=7; \end{array}\right\}$$

Research Methodology

To improve WAV file encryption security, a multilayer method is employed in this study. **Safeguards against potential cryptographic vulnerabilities without compromising the distinctive characteristics of the audio data by embracing a multilayer method**. The multilayer method employs skilled security measures in audio encryption and decryption [18]. The robust cryptographic key is produced by the method for encrypting WAV files by using the PBKDF2 HMAC SHA256 algorithm. There are various benefits of SHA256, including similar strength, stability and fixed-size output, which in turn knowingly improves key security. The ChaCha20 encryption algorithm is flawlessly integrated into the proven methodology. This method includes audio encryption in the method of encryption, increasing its defense against threats to security with its logical basis. ChaCha20 operates as a symmetric key algorithm, selected for its speed and efficiency, working on a stream cipher basis. Potential threats are therefore avoided by adding an extra barrier that enhances the overall resilience of the audio encryption process.

Therefore, the unified approach, which includes custom encryption, PBKDF2-HMAC SHA256 key derivation, and ChaCha20 encryption, has proven to be a strong and flexible security framework for audio encryption. The above-described process of encryption is shown in Fig. 2.

Design Insights: Integrating ChaCha20 with Non-Linear Transformations

ChaCha20 was chosen as the cryptographic method for this study because of its robust security, computational efficiency, and resistance to side-channel attacks. Unlike the Advanced Encryption Standard (AES), which is based on sophisticated substitution-permutation operations and S-boxes, ChaCha20 is a stream cipher that uses basic

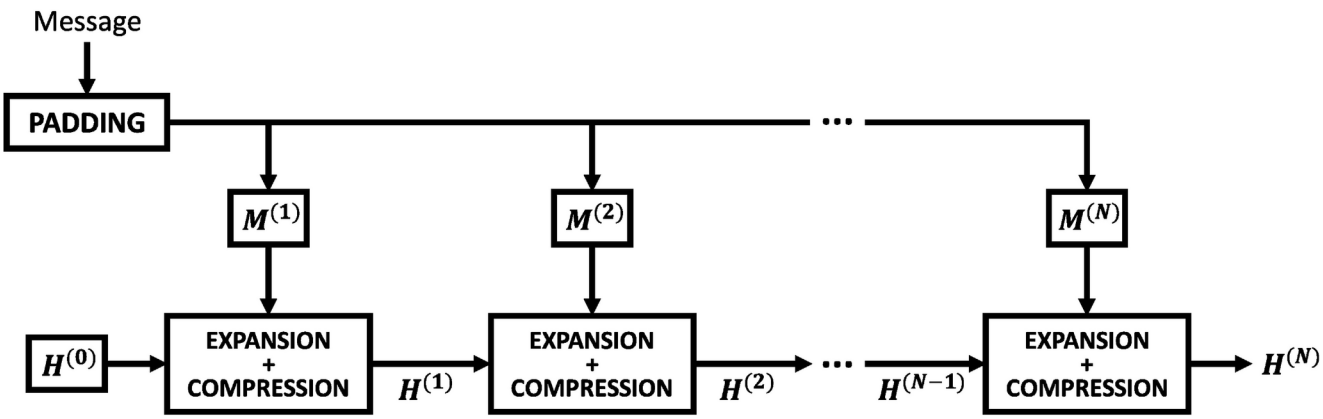


Fig. 1 Diagrammatic representation of SHA-256 methodology

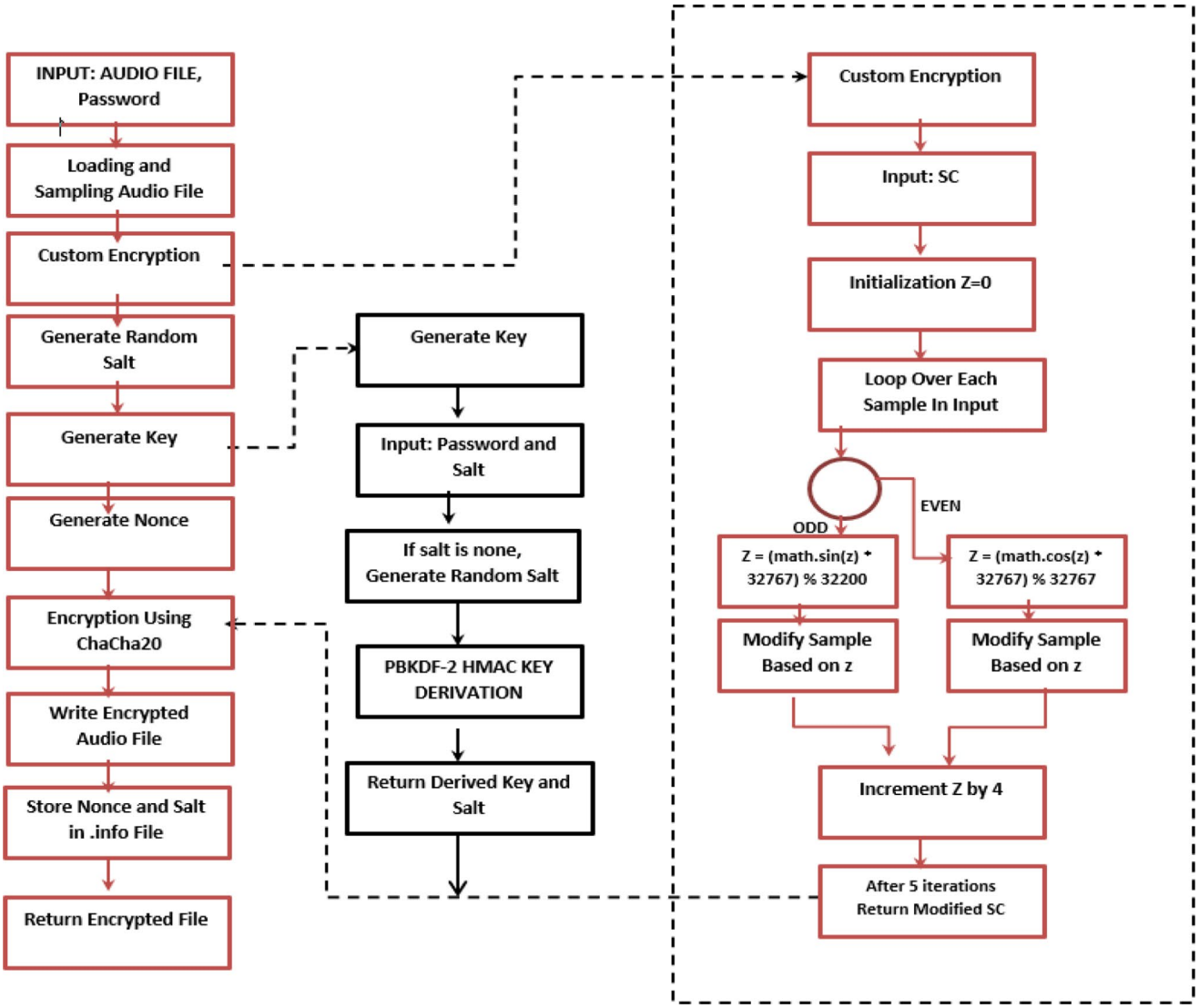


Fig. 2 Block Diagram of the Proposed Encryption Scheme

bitwise operations like XOR, addition, and rotation. ChaCha20's wide key space of 2^{256} makes it resilient to brute-force assaults and ensures uniform distribution of encrypted data. In terms of audio encryption, ChaCha20 has specific advantages over AES since it runs in a continuous stream rather than fixed-size blocks.

Role of Non-Linear Functions in Enhancing Encryption Security

Using sine and cosine-based nonlinear functions prior to the use of the ChaCha20 algorithm, the new encryption method enhances security. Such modifications encompass uneven modifications to the encryption process, which enhance resistance to statistical and differential cryptanalysis through enhanced confusion and diffusion. Through realignment of the frequency spectrum of the input data, such functions render it far more difficult to reconstitute the original signal without the decryption key [26]. Instead of direct encryption, the technique adopted in the proposal involves several rounds of non-linear transformations before ChaCha20, which largely alters the correlation between input and encrypted audio signals. This gives high entropy and a low correlation coefficient, hence strengthening encryption. This multi-layer approach strengthens the security of the whole encryption scheme and renders it stronger cryptographically resistant to attacks. Many research studies support these conclusions, such as Zhao et al. (2021), who looked into chaotic systems and sine-cosine functions for robust multimedia encryption [26], and Zhang et al. (2021), who examined non-linear sine-cosine transformations for improved audio encryption [25]. Al-Rimy et al. (2020) performed a comparative analysis of sine and cosine-based

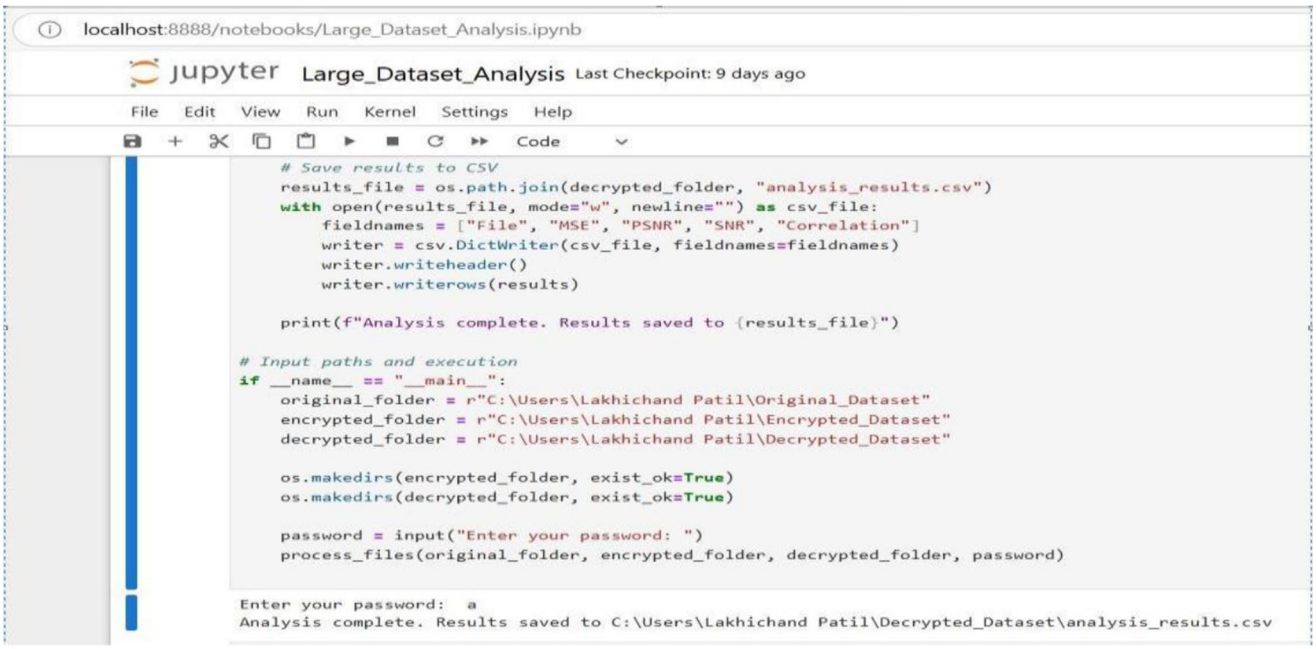


Fig. 3 Dataset Encryption and Metrics Analysis Framework

non-linear cryptography algorithms for audio encryption [28], while Jafari and Mousavi (2020) concentrated on the use of non-linear functions for cryptographic key generation in secure audio transmission [27].

Block Diagram of Encryption Process

To ensure security and efficiency, the encryption process is structured, with multiple layers. It starts with the preprocessing stage, in which the incoming audio stream is normalized and sampled with Librosa to ensure homogeneity. A random nonce and salt are created to ensure that each encryption session is distinct. During the key generation PBKDF2-HMAC-SHA256 is used to generate a cryptographic key using the user's password. The resulting key is used for encryption process. The multi-round non-linear transformation stage next applies many iterations of sine and cosine functions to the audio data, raising entropy and decreasing correlation, making it more resistant to cryptanalysis. The altered data is then encrypted with the ChaCha20 stream cipher, where the resulting key and nonce ensure that even identical plaintext inputs create distinct ciphertexts. The encrypted audio is then saved along with metadata like the nonce and salt for decryption. During the decryption procedure, the receiver uses metadata. ChaCha20 decryption is used, followed by inverse non-linear transformations to successfully recreate the original signal.

Proposed Audio Encryption Scheme

Input:

- *audio_file* ← WAV Audio File
- *K*1 ← Derived Key from Password.
- *password* ← Password for Key Generation.

Output:

- *Encrypted_audio* ← Encrypted WAV File.

Initialize:

- *rounds* ← 5.
- *salt* ← Random 16-byte salt (or provided salt if available).
- *nonce* ← Random 16-byte nonce.
- *key* ← Derived 256-bit key generated from password and salt using PBKDF2HMAC (SHA-256, 10 iterations).
- *Cipher* ← ChaCha20 cipher initialized with *key* and *nonce*.

1. Audio Preprocessing:
   Load the audio file into an array x with sample rate sr
   Normalize audio data to 16-bit integer range, converting x to signed integers array *Sc*
2. Custom Encryption Rounds:

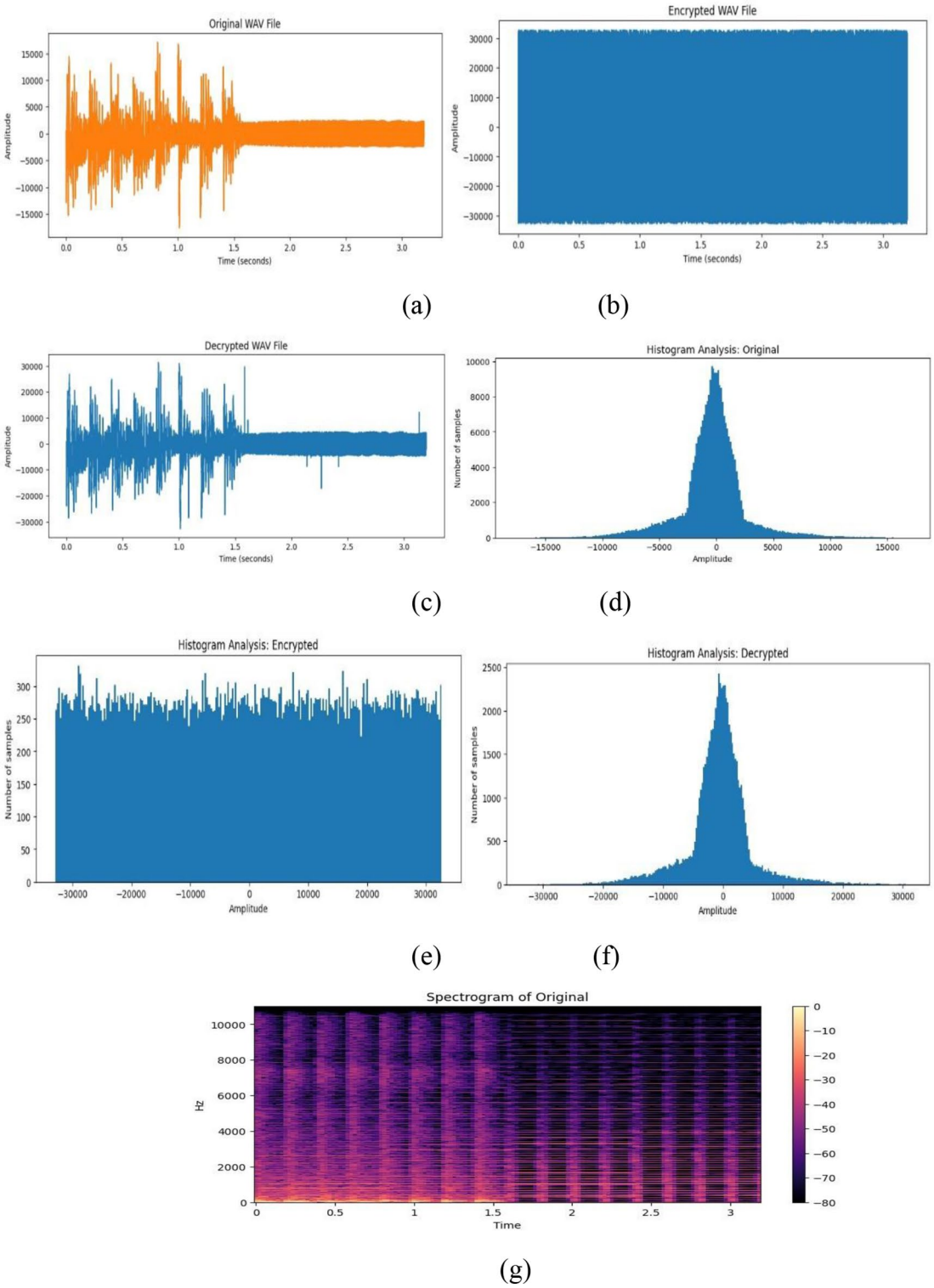


Fig. 4 Audio file 1 (**a**) Source WAV file- which is original file (**b**) Output WAV file- which is encrypted file (**c**) Decrypted WAV file (**d**) Histogram analysis: Original (**e**) Histogram analysis: Encrypted (**f**) Histogram analysis: Decrypted (**g**) Spectrogram of Original

For each *r* in round *s*:

- Set $z \leftarrow 0$.
- For each sample *i* min *sc*
  - If *i* is odd, set $z \leftarrow (\sin(z)*36767)\%32200$ and update *sc[i]*← *sc[i]*-z.
  - If i is even, set $z \leftarrow (\cos(z)*36767)\%32767$ and update *sc[i]*← *sc[i]*-*z*
  - Increment x by 4 after each sample.

3. Key Generation and Cipher Initialization:
   Generate derived 256-bit key, K1, from password and salt using PBKDF2HMAC
   Initialize ChaCha20 cipher with derived key, K1, and nonc
4. Encryption:
   Encrypt sc by passing it through the ChaCha20 cipher
   Store encrypted data in Encrypted_audio
5. Save Encrypted Audio:
   Save Encrypted_audio in a new WAV file with modified file name and Store Store nonce and salt in an associated.info file

Securing Audio Data: Challenges, Objectives and Goals

Audio files are susceptible to a variety of attacks, including brute force, replay, differential cryptanalysis, and plaintext attacks, due to their steady signals and high redundancy. Such vulnerabilities make it very hard to protect sensitive audio information, particularly in places where secure transmission and storage of audio is a requirement—e.g., secure VoIP calls, healthcare applications, voice memos in confidence, forensic protection of audio, and privacy-guaranteeing voice assistants. Combining non-linear sine and cosine transformations with the ChaCha20 stream cipher, the solution ensures high confusion and diffusion and thus is highly resistant to common attacks such as statistical, brute-force, and differential attacks. Experimental results for over 480 audio samples were consistently high in entropy values, low in correlation coefficients, and nearly zero Mean Squared Error (MSE) between the decrypted and original signals, ensuring effective security without perceptual degradation. The objective of this project is to enhance the integrity and reliability of audio data so that it may be utilized in numerous sensitive applications.

Novelty and Advancements in Audio Encryption

By combining a stream cipher with multi-round non-linear processing, the suggested encryption methodology sets itself apart from conventional audio encryption methods. The suggested method adds an extra layer of security by applying sine and cosine-based modifications before encryption, in contrast to traditional methods that mostly rely on block ciphers like AES and DES. After these changes, ChaCha20 is used to increase the encryption's resilience even more, creating a two-layer process that is absent from traditional techniques. According to their correlation coefficients, traditional encryption methods frequently show statistical similarities between plaintext and ciphertext. On the other hand, the suggested method increases resistance to statistical attacks by drastically lowering the correlation coefficient (~0.000118). Additionally, the method's efficacy is confirmed by cryptanalysis measures, with UACI (~0.33) and NPCR (~99.99%) values showing high sensitivity to plaintext modifications and guaranteeing a noticeable avalanche effect, which guards against differential attacks. Furthermore, the technique ensures near-perfect unpredictability in encrypted data by achieving high entropy (~7.9999 bits). Its effectiveness in controlling encrypted file sizes is yet another significant benefit. The suggested approach maintains or even decreases file size through optimized transformations, making it both computationally efficient and storage-friendly in contrast to typical encryption approaches that introduce significant file size growth owing to padding. Comparative outcomes using single-stage encryption (ChaCha20 alone or substitution-permutation alone) are that the two-stage method shows greater entropy, lower correlation coefficients, and greater resistance to cryptanalytic attacks.

Result Analysis

Both the encrypted and decrypted datasets' quality and security can be determined with a Python-program called the Dataset Encryption and Metrics Analysis Framework. 480 WAV files are treated in an attempt to gain correlation coefficients, mean squared error (MSE), peak signal-to-noise ratio (PSNR), signal-to-noise ratio (SNR), and more metrics. The framework saves outputs to a CSV file for additional analysis, automates result production, and guarantees smooth directory management. For cryptographic research and performance analysis, this tool is perfect.

Four different types of audio files are considered for analysis. Figures 5, 6 and 7, and Fig. 8 give the histogram and spectrogram analysis of these audio files. The original audio files are given in subdivision (a) of these figures,

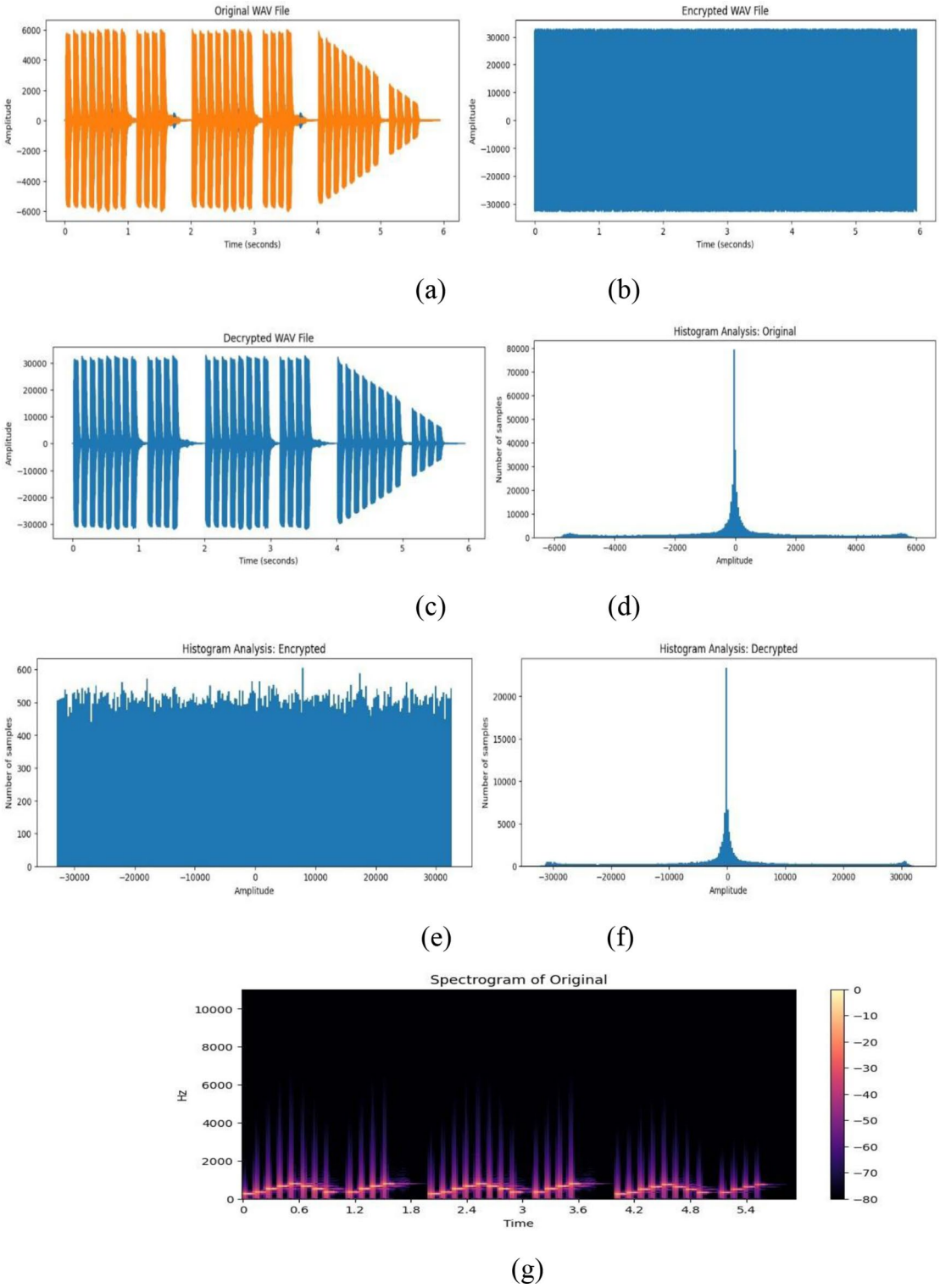


Fig. 5 Audio file 2 (**a**) Source WAV file- which is original file (**b**) Output WAV file- which is encrypted file (**c**) Decrypted WAV file (**d**) Histogram analysis: Original (**e**) Histogram analysis: Encrypted (**f**) Histogram analysis: Decrypted (**g**) Spectrogram of Original

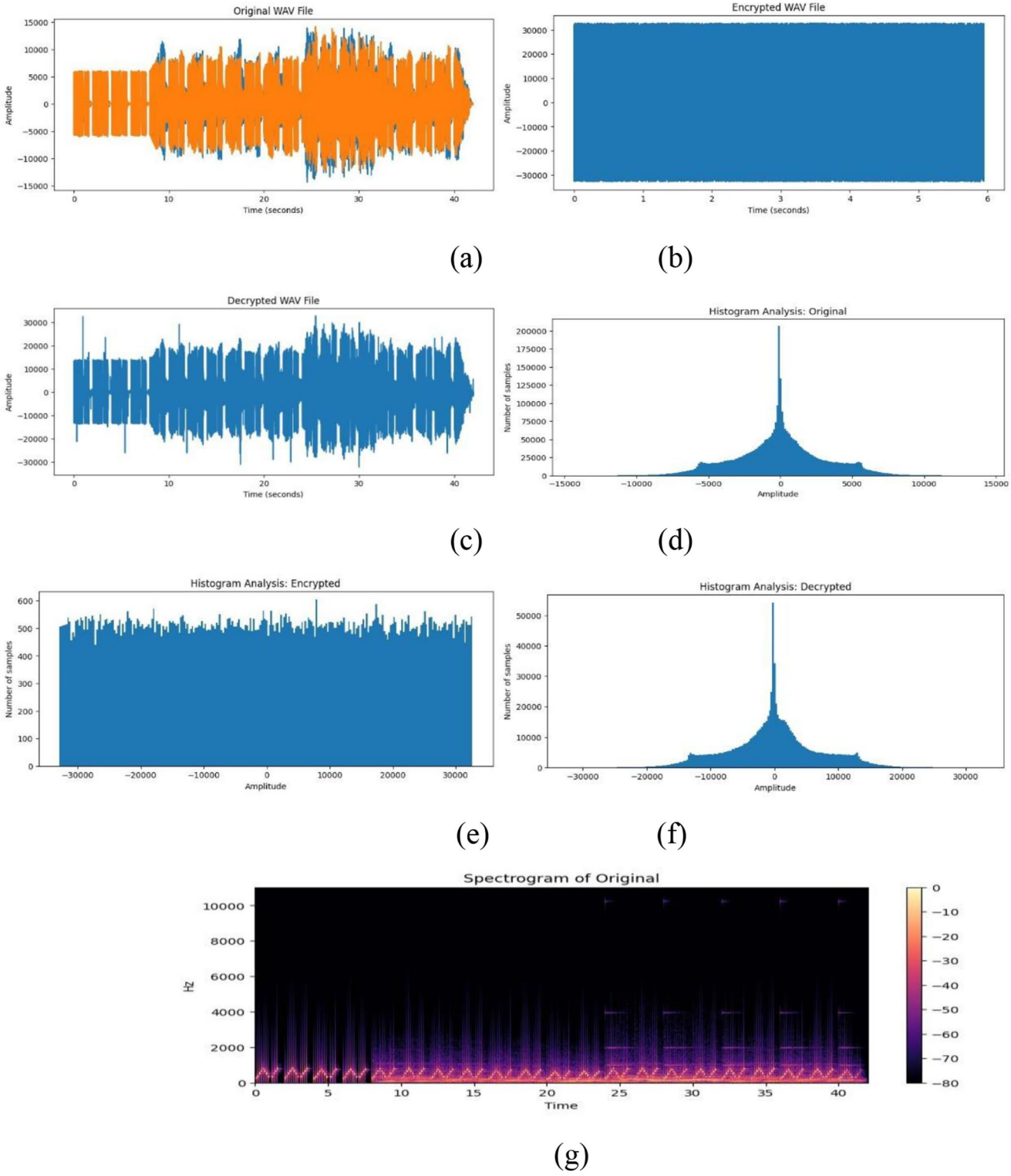


Fig. 6 Audio file 3 (**a**) Source WAV file- which is original file (**b**) Output WAV file- which is encrypted file (**c**) Decrypted WAV file (**d**) Histogram analysis: Original (**e**) Histogram analysis: Encrypted (**f**) Histogram analysis: Decrypted (**g**) Spectrogram of Original

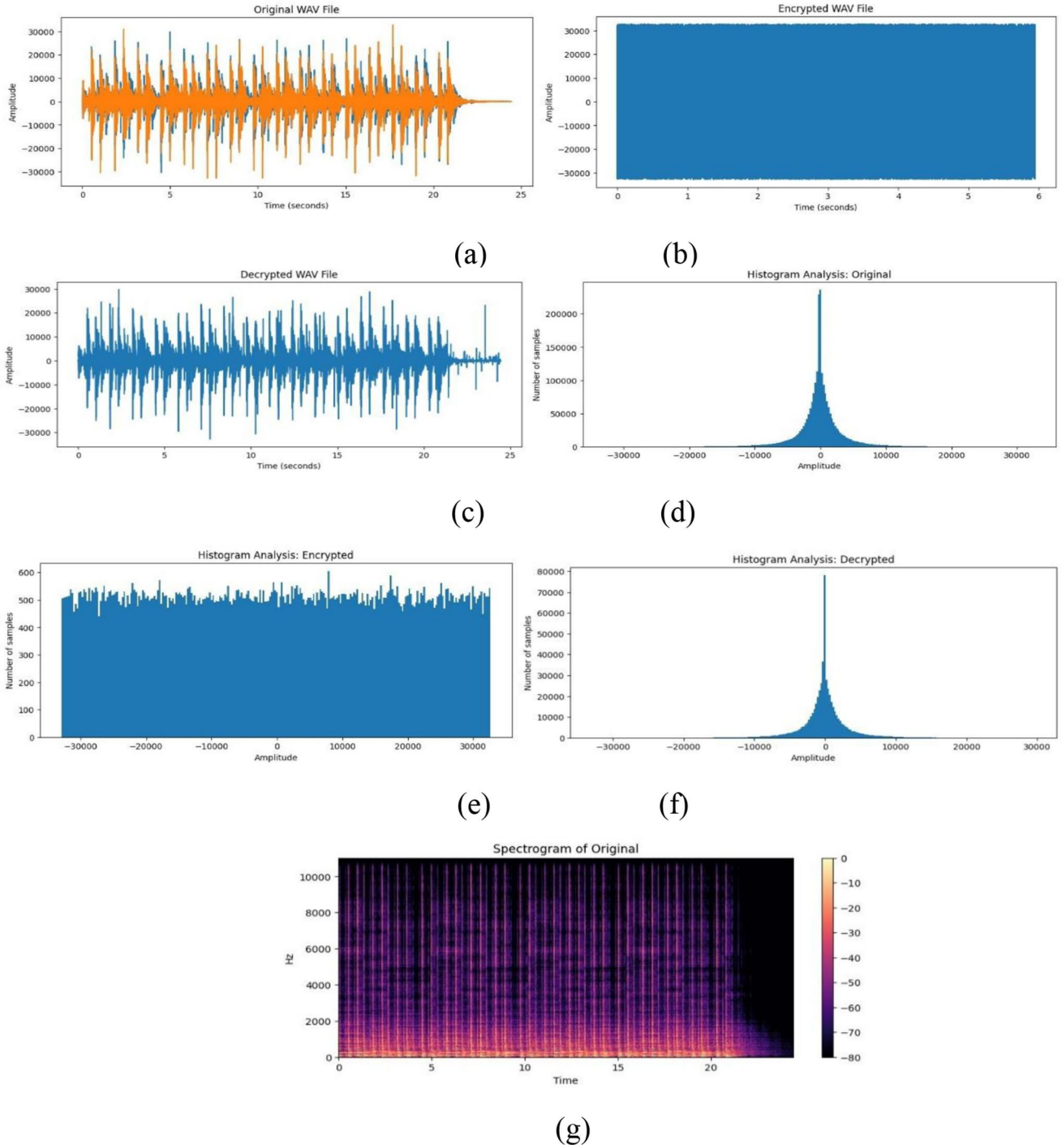


Fig. 7 Audio file 4 (**a**) Source WAV file- which is original file (**b**) Output WAV file- which is encrypted file (**c**) Decrypted WAV file (**d**) Histogram analysis: Original (**e**) Histogram analysis: Encrypted (**f**) Histogram analysis: Decrypted (**g**) Spectrogram of Original

encrypted audio files in subdivision (b) of these figures, decrypted audio files in subdivision (c) of these figures, original histogram analysis in subdivision (d) of these figures, encrypted histogram analysis in subdivision (e) of these figures, decrypted histogram analysis in subdivision (f) of these figures and spectrogram analysis in subdivision (g) of Figs. 5–8.

Histogram Analysis

A precise parameter for assessing the effectiveness of an encrypted audio transmission is histogram analysis. The original audio file should be encrypted using a reliable encryption algorithm to transform it into an approximately consistent sample value distribution of random-like sound. Therefore making it vulnerable to statistical attacks.

Spectrogram Analysis

A spectrogram visually represents the range of frequencies in an audio file, depicting how it evolves over time. It is derived using the Fourier transform, illustrating both time and frequency on a two-dimensional graph. The Fourier transform is used to split the audio recordings into smaller segments and determine the frequency distribution magnitude for each segment.

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Correlation Analysis

Correlation analysis, a statistical indicator, may assess how resilient an encryption technique is to different statistical attacks. In essence, it calculates the correlation between corresponding parts of the source audio and the resulting encrypted audio files. To do this, the audio file is transformed into a noisy, random signal with low correlation coefficient using a strong encryption method. The correlation coefficient can be determined using the formula provided below:

$$r_{xy} = \frac{\operatorname{cov}(x,y)}{\sigma_x \sigma_y} = \frac{\frac{1}{N_s}\sum_{i=1}^{N_s}(x_i - E(x))(y_i - E(y))}{\sqrt{\frac{1}{N_s}\sum_{i=1}^{N_s}(x_i - E(x))^2}\sqrt{\frac{1}{N_s}\sum_{i=1}^{N_s}(y_i - E(y))^2}}$$

and $\sigma_x, \sigma_y \neq 0$

In the equation given above, covariance is indicated by cov(x, y) among all the files. In the given equation, xi and yi show the standard deviations of files, the source audio file and encrypted audio file both, while E(x) and E(y) represent sample means. Sample values for source and encoded audio files are shown as xi and yi respectively. Overall number of samples is presented with Ns. Table 1 presents the comparison of the correlation coefficients of the five audio files. The results show that the correlation coefficient values are very close to zero, which means that there is no correlation between the dual files and shows the randomness of the encrypted audio file. This issue demonstrates that the encryption technique that has been proposed is of a very high quality. **When an attacker picks up the encrypted audio, this near-zero correlation provides them with no indication of the original signal even if they possess part of the original audio (i.e., known-plaintext attack contexts).**

Table 1 Using PSNR, correlation coefficient, and SNR to compare the encrypted output audio files with the source WAV files [5]

WAV files	Size	Duration (Sec.)	Correlation coefficient	SNR	PSNR
Sample-3s	2.15 MB	3.195646	**0.00226**	-13.587	-1.0099
Sample-3ss	859 KB	3.195646	**-0.00119**	-13.578	-1.0897
S	808 KB	3.195646	**0.000119**	-13.586	-1.0979
Sample-9s	830 KB	9.586939	**-0.000148**	-18.550	-0.915
Sample-12 S	652 KB	12.120852	**0.002433**	-21.136	-8.5836

Table 2 Derived different passwords and resulting keys

Derived Password	Key
user_password	d230414c07b114f12b9c8aa41bc80a83eaaf-8f141f1b59bae04e3fb7a6b821f9
User_password	7e417b46751da886c851e11de5b4f-77057654cedb1e0267bbc4f524e05ba3e20
user_password123	244c382d379301dc325da6a36d19fd89c74ff-cef5f41ca1d5269c8f35ab1640d
password123_user	78b9f689c8a6dd5520d3a736e0d-4932ca7b740d0d27169b56f763051391cacd8

Key Sensitivity

A strong audio encryption method needs to react very quickly to a slight alteration in its keys. This implies that a very small alteration to the key value must result in an entirely new encrypted audio file [5].

Every password variation yields a unique derived key, suggesting that minor modifications to the password cause major modifications to the derived keys. This demonstrates that the key derivation process is sensitive to variations in the input parameters, which is desirable for cryptographic security. Table 2 depicts derived different passwords and resulting keys. Each generated key is unique and exhibits no clear patterns or repetitions, which is important for ensuring the security of the encryption process.

NSCR and UACI

One of the most important measures of an encryption scheme's strength is its resistance to differential attacks. The NSCR stands for-Number of Sample Change Rate [17] whereas UACI stands for-Universal Average Changing Intensity [17] tests are utilized to ascertain the level of resistance. The application of these tests requires the encryption of two audio segments that differ from one another only by a single sample using the equivalent key. After that, the two encoded audio elements are given comparison using NSCR and UACI, which are computed subsequently: [21, 22, 23].

$$\begin{aligned} \text{NSCR} &= \sum_i \frac{D_i}{N_s} \times 100\% \; i = 1,2,\ldots\ldots\ldots,N \\ UACI &= \frac{1}{N_s}\left[\frac{\sum_i x_i - x_i'}{65535}\right] \\ \text{Where } D_i &= \begin{cases} 1, & x_i \neq x_i' \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

xi and xi' are samples taken at the ith position of the encrypted WAV file whereas Ns indicates the length of the segment. The comparison of encrypted and decrypted WAV files based on UACI and NSCR metrics is represented in Table 3.

The suggested audio encryption technique realizes a low UACI of 0.3321%, which is very important in audio encryption because it proves that slight modifications in the original audio result in only minor variations in the encrypted audio. This is essential in real-time audio systems where imperceptible variations may arise due to compression, noise, or transmission instability. A low UACI guarantees that the encryption is stable under such conditions, maintaining the quality and integrity of the decrypted audio. At the same time, a high NSCR of 99.99% guarantees good diffusion, ensuring good security against differential and statistical attacks. This balance guarantees both reliable transmission and good audio content protection.

Signal to Noise Ratio

For the purpose of determining the quality of the signal, the Signal to Noise Ratio (SNR) is utilized. Since the readings are more than 0 decibels, it can be concluded that the signal is greater than the noise. Taking into account both the encrypted and the original audio files, the signal-to-noise ratio (SNR) could be determined as follows:

$$SNR = 10^{*}\log_{10}\frac{\sum_{i=1}^{N_s} x_i^2}{\sum_{i=1}^{N_s}(x_i - y_i)^2}$$

where Ns is the quantity of samples and x_i and y_i are the original samples and encrypted samples.

Table 3 Represents WAV Files comparison i.e. File to be encrypted and the output (encrypted) File, focusing on UACI & NSCR metrics

Audio File	NSCR	UACI
Sample-3s	99.9985%	0.3321%
Sample-3ss	99.9985%	0.3325%
S	100.0%	0.3322%
Sample-9s	99.9992%	0.3322%
Sample-12 S	99.9989%	0.3325%

Table 4 Comparing the source audio files with encrypted output WAV files using RMS and CF

Audio File	Encrypted Audio RMS	Encrypted Audio CF
Sample-3s	0.5626	1.73
Sample-3ss	0.5625	1.73
s	0.5626	1.73
Sample-9s	0.5626	1.73
Sample-12 S	0.5626	1.73

Peak Signal to Noise Ratio

The following expression can be utilized in order to obtain the mean squared error for two streams that are contained in vectors X and Y [5].

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(X[i] - Y[i])^2$$

The PSNR may be computed as follows, X represents the source audio file and Y represents the encrypted output audio file [5].

$PSNR = 10^{*}\log_{10}\left(\frac{MAX^2}{MSE}\right)$ where MAX is the stream's maximum value.

Values of the Root Mean Square (RMS) & Crest Factor (CF)

To calculate root means square measurement for a sound signal, consider the volume level. The following diagram illustrates how the input signal's computation is represented, even if its mean is 0 as it resembles the standard deviation.

$$RMS = \sqrt{\frac{1}{N}\sum_{i=1}^{N}|A_i|^2}$$

A waveform characteristic called the crest factor (CF) is obtained by performing the division operation where the peak values as numerators and the effective values are as denominators. This ratio is used to determine the crest factor. This number is the smallest measurement needed to capture a waveform's peak extremes. Table 4 shows a comparison between the source WAV file and the encrypted output WAV file using RMS and CF. A CF ratio of 0 dB shows

Table 5 Analyzing the suggested approach compared to current methods identified in the literature

Algorithm used	Correlation Coefficient	Key Space	SNR	PSNR	UACI	NSCR	RMS	CF
[14]	0.0263	-	-	9.53				
[15]	0.0014	2^{128}	-1.47					
[16]	0.00035	2^{128}	-	4.426		99.997		4.8
[18]	0.0092	2^{144}						
[19]	0.0038	2^{149}	-16.04	4.39		99.996		4.8
[20]	0.000207	2^{256}				99.9989	0.6	4.8
AES	0.0097	2^{256}	-1.4		33.32	99.603		
3DES	0.1704	2^{168}	-0.25		33.34	99.59		
Proposed	**0.000119**	$\mathbf{2^{256}}$	**-13.587**	**-1.0099**	**33.21**	**99.9985**	**0.5**	**1.73**

that there are noeaks in the DC current. A higher CF indicates the presence of peaks. In the following, it is presented:

$$CF = 20\log_{10}\frac{|V_{\text{Peak}}|}{V_{RMS}}$$

As can be seen, the encrypted audio files' RMS and CF values are nearly 0.56 and 1.73, respectively. This demonstrates that the matching encrypted audio files and the original audio files have no statistical link.

Achieving High Entropy for Robust Audio Encryption

Entropy is a key measure of randomness, crucial in evaluating encryption effectiveness, especially for audio files. It quantifies how evenly the bits in an encrypted file are distributed, with higher values indicating greater randomness and stronger security. Entropy is calculated using Shannon's Information Theory and the probability that each symbol will appear. The highest possible entropy for an 8-bit audio file is 8 bits, which indicates optimal randomness. An encrypted file that has a high entropy value near 8 is difficult to cryptanalysis because it lacks predictable patterns. After encryption the entropy increases to 7.999 bits, showing maximum unpredictability, the correlation factor falls to $r = 0.0001$, showing no nonlinear link. These results indicating that the sound was properly randomized after the encryption process. High entropy, as 7.999 bits, is required for encryption audio data to remain safe and secrets. In contrast, low entropy shows low randomness, considering the file at risk. For instance, the initial audio represents regular patterns having an entropy of 4.5 bits and a correlation coefficient of $r = 1$.

Comparative Assessment and Trade-off Analysis

This section compares the proposed method with conventional methods based on standard security parameters (Table 5). The proposed method demonstrates a low correlation coefficient, which means that the original and encrypted audio have little similarity. The strong resistance against brute-force attacks is guaranteed by having a large key space (2^{256}), and a lower SNR indicates strong encryption. The approach outperforms in sensitivity to keys, entropy, and attack resistance due to the incorporation of non-linear operations and ChaCha20, which improve diffusion and randomness. Although it has slightly more computational expense, it is still efficient and can be handled, thus it is a prime contender for secure audio usage.

Key Advancements in Audio Encryption

This study provides a novel method for encrypting audio data by integrating modern cryptography techniques with trusted evaluation methods. Higher-speed and secure encryption of audio data is made possible by the encryption process incorporation of PBKDF2-HMAC with SHA-256 for key derivation, which validates resistance to dictionary and brute-force attacks, and ChaCha20, an effective and secure stream cipher. In order to provide resulting key and enhance unpredictability through cycles of non-linear functions like sine and cosine, the suggested method includes a special encryption logic that manages salt, and nonce. To evaluate the quality of the encryption, metrics including correlation coefficients, mean squared error (MSE), peak signal-to-noise ratio (PSNR), and signal-to-noise ratio (SNR) are examined. This complete evaluation is rarely observed in audio encryption research. Also, the dual-stage encryption approach, which combines ChaCha20 with the unique encryption logic, provides a strong security model that effectively mitigates common cryptographic attacks. The innovation and effectiveness of the proposed approach are highlighted by comparisons with existing methods,

representing higher performance in metrics such as entropy, Number of Pixels Change Rate (NPCR), and, Unified Average Changing Intensity (UACI).

Conclusion

The proposed study offers a strong and effective framework for audio data encryption, integrating current unique logic with current cryptographic algorithms to address important issues in audio security. By employing PBKDF2-HMAC with SHA-256 for secure key derivation and ChaCha20 as the primary encryption method, the framework maintains computational efficiency while safeguarding against dictionary and brute-force attacks. The integration of nonlinear functions, such as cosine and sine, into the encryption rounds improves the diffusion and confusion properties, significantly improving the security of audio data.

The significance of the suggested method is confirmed by its higher performance in key cryptanalysis metrics compared to other methods. The near-ideal entropy of **7.9999** bits represents maximum unpredictability in the encrypted file. Strong encryption requires high sensitivity to changes in primary file, as proved by the **99.99%** NPCR and **0.33%** UACI. Additionally, the strength of the proposed structure is reinforced by a correlation coefficient of **0.000119**, indicating a very weak correlation between the original and encrypted file.

These results indicate the efforts made to ensure that the encryption procedure not only satisfies but also exceeds the most recent requirements for the security of audio data. The dual-stage encryption procedure creates a strong safety framework that successfully counters typical cryptographic risks, as proven by rigorous cryptanalysis.

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